# Experiment 2
# Perform Operations on Vectors, Matrices and Arrays in R

## PART B

PART B: TO BE COMPLETED BY STUDENTS)

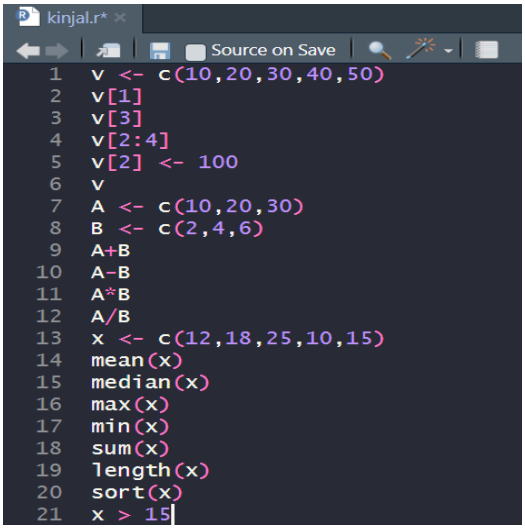
***(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)***

| | |
|---|---|
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| Class : S.E. - C | Batch : C-2 |
| Date of Experiment: | Date of Submission:- |
| Grade : | |

**Observation: (Run the programs given above in R studio and paste screenshots of along with the output)**

1. **Vector Operations**

**Input :-**



**Output:-**

```
> v <- c(10,20,30,40,50)
> v[1]
[1] 10
> v[3]
[1] 30
> v[2:4]
[1] 20 30 40
> v[2] <- 100
> v
[1]  10 100  30  40  50
> A <- c(10,20,30)
> B <- c(2,4,6)
> A+B
[1] 12 24 36
> A-B
[1]  8 16 24
> A*B
[1]  20  80 180
> A/B
[1] 5 5 5
> x <- c(12,18,25,10,15)
> mean(x)
[1] 16
> median(x)
[1] 15
> max(x)
[1] 25
> min(x)
[1] 10
> sum(x)
[1] 80
```

```
> length(x)
[1] 5
> sort(x)
[1] 10 12 15 18 25
> x > 15
[1] FALSE  TRUE  TRUE FALSE FALSE
>
```

2. **Matrix Operations**
   **Input:-**

kinjal.r

Source on Save

```
1  A <- matrix(c(1,2,3,4,5,6),nrow=2,ncol=3)
2  A
3  B <- matrix(c(1,2,3,4,5,6),nrow=2,byrow=TRUE)
4  A[1,2]
5  A[2,3]
6  A[1,]
7  A[,2]
8  A <- matrix(c(1,2,3,4),2,2)
9  B <- matrix(c(5,6,7,8),2,2)
10 A+B
11 A-B
12 A*B
13 A%*%B
14 t(A)
15 det(A)
16 solve(A)
17 rowSums(A)
18 colSums(A)
19 diag(A)
```

**Output:-**

```
> A <- matrix(c(1,2,3,4,5,6),nrow=2,ncol=3)
> A
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
> B <- matrix(c(1,2,3,4,5,6),nrow=2,byrow=TRUE)
> A[1,2]
[1] 3
> A[2,3]
[1] 6
> A[1,]
[1] 1 3 5
> A[,2]
[1] 3 4
> A <- matrix(c(1,2,3,4),2,2)
> B <- matrix(c(5,6,7,8),2,2)
> A+B
     [,1] [,2]
[1,]    6   10
[2,]    8   12
> A-B
     [,1] [,2]
[1,]   -4   -4
[2,]   -4   -4
> A*B
     [,1] [,2]
[1,]    5   21
[2,]   12   32
```

```
> A%*%B
     [,1] [,2]
[1,]   23   31
[2,]   34   46
> t(A)
     [,1] [,2]
[1,]    1    2
[2,]    3    4
> det(A)
[1] -2
> solve(A)
     [,1] [,2]
[1,]   -2  1.5
[2,]    1 -0.5
> rowSums(A)
[1] 4 6
> colSums(A)
[1] 3 7
> diag(A)
[1] 1 4
>
```

3. **Array Operations**
   **Input:-**

```r
A <- array(1:24,dim=c(3,4,2))
A[2,3,1]

A[1,,2]

A[,,1]
A1 <- array(1:12,dim=c(2,3,2))

A2 <- array(13:24,dim=c(2,3,2))

A1+A2

A1*A2
dim(A)
length(A)
apply(A,c(1),sum)
apply(A,c(2),mean)
```

Output:-

```
> A <- array(1:24,dim=c(3,4,2))
> A[2,3,1]
[1] 8
> A[1,,2]
[1] 13 16 19 22
> A[,,1]
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12
> A1 <- array(1:12,dim=c(2,3,2))
> A2 <- array(13:24,dim=c(2,3,2))
> A1+A2
, , 1

     [,1] [,2] [,3]
[1,]   14   18   22
[2,]   16   20   24

, , 2

     [,1] [,2] [,3]
[1,]   26   30   34
[2,]   28   32   36
```

```
> A1*A2
, , 1

     [,1] [,2] [,3]
[1,]   13   45   85
[2,]   28   64  108

, , 2

     [,1] [,2] [,3]
[1,]  133  189  253
[2,]  160  220  288

> dim(A)
[1] 3 4 2
> length(A)
[1] 24
> apply(A,c(1),sum)
[1]  92 100 108
> apply(A,c(2),mean)
[1]  8 11 14 17
>
```

**Observation Table:**

| Operation | Result |
|---|---|
| Vector Creation | Successfully created one-dimensional vectors |
| Vector Arithmetic | Addition, subtraction, multiplication, and division performed correctly |

| Operation | Result |
|---|---|
| Statistical Functions | Mean, median, sum, maximum, and minimum computed successfully |
| Matrix Operations | Matrix addition, multiplication, transpose, determinant, and inverse executed correctly |
| Array Operations | Three-dimensional arrays created and manipulated successfully |
| Apply Function | Row-wise and column-wise computations performed successfully |

**Result**

The experiment was successfully performed. Vectors, matrices, and arrays were created using R, and various arithmetic, statistical, and matrix operations were executed correctly. The experiment demonstrated efficient manipulation of one-dimensional, two-dimensional, and multidimensional data structures in R.

**Applications**

- Statistical data analysis
- Scientific computing
- Machine Learning preprocessing
- Image processing (multidimensional arrays)
- Financial data analysis
- Engineering computations
- Healthcare analytics
- Artificial Intelligence applications

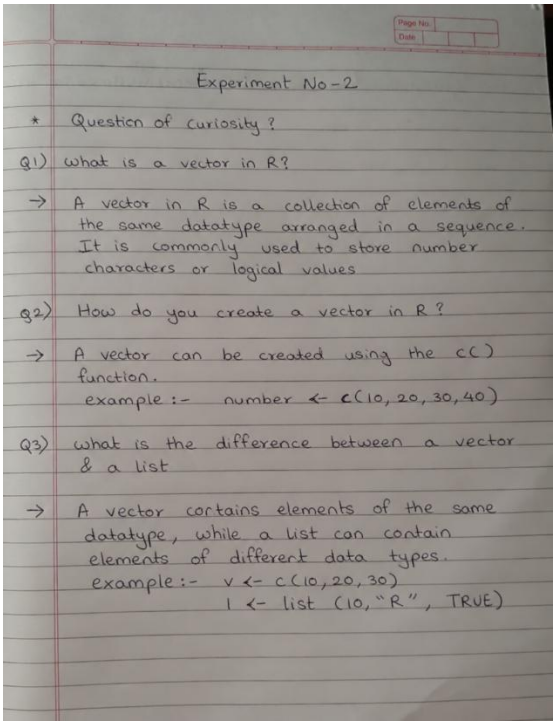
**Precautions**

1. All elements in a vector or matrix must be of the same data type.
2. Ensure matrices have compatible dimensions before performing matrix multiplication.
3. The inverse of a matrix exists only if its determinant is non-zero.
4. Use proper indexing while accessing vector, matrix, or array elements.
5. Verify dimensions before performing arithmetic operations on arrays.
6. Save the R script before execution to avoid loss of work.

**Questions of Curiosity**

1. What is a vector in R?
2. How do you create a vector in R?
3. What is the difference between a vector and a list?
4. What are the different types of vectors?
5. How is a matrix different from a vector?

**Answers:-**



Experiment No-2

* Question of curiosity ?

Q1) what is a vector in R?

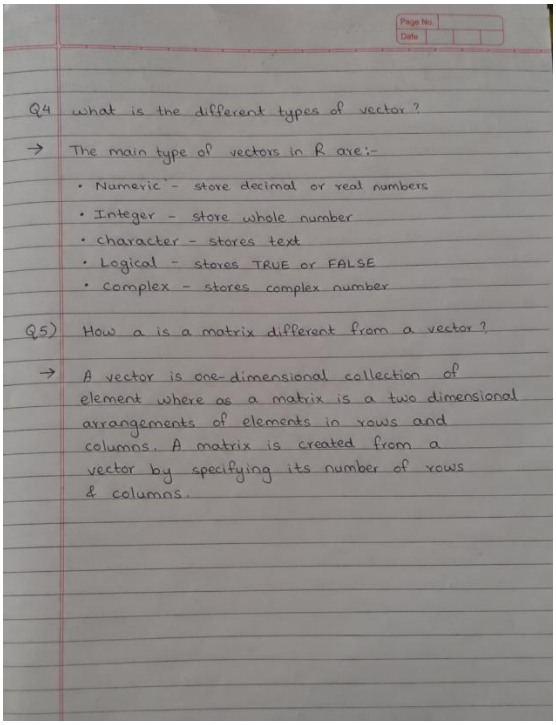
→ A vector in R is a collection of elements of the same datatype arranged in a sequence. It is commonly used to store number characters or logical values

Q2) How do you create a vector in R ?

→ A vector can be created using the c() function.
example :- number <- c(10, 20, 30, 40)

Q3) what is the difference between a vector & a list

→ A vector contains elements of the same datatype, while a list can contain elements of different data types.
example :- v <- c(10, 20, 30)
l <- list (10, "R", TRUE)



Q4 what is the different types of vector ?

→ The main type of vectors in R are:-

- Numeric - store decimal or real numbers
- Integer - store whole number
- character - stores text
- Logical - stores TRUE or FALSE
- complex - stores complex number

Q5) How a is a matrix different from a vector ?

→ A vector is one-dimensional collection of element where as a matrix is a two dimensional arrangements of elements in rows and columns. A matrix is created from a vector by specifying its number of rows & columns.

**Viva Voce Questions**

6. What is a vector in R?
7. How do you create a vector in R?
8. What is the difference between a vector and a list?
9. What are the different types of vectors?
10. How is a matrix different from a vector?
11. Explain the `matrix()` function.
12. What is the purpose of `%*%` in R?
13. Differentiate between `*` and `%*%`.
14. What is the transpose of a matrix?
15. What is the determinant of a matrix?
16. When can the inverse of a matrix be calculated?
17. What is an array in R?
18. How is an array different from a matrix?
19. Explain the `apply()` function.
20. What is the purpose of `dim()` and `length()` functions?
21. How do you access elements in a three-dimensional array?
22. What statistical functions are commonly used with vectors?
23. What happens if vectors of unequal lengths are added in R?

24. What are the applications of matrices in data science?
25. Why are vectors, matrices, and arrays important in R programming?